

properties quickly and accurately.

Electrical characterisation is essential to gain insight into phenomena that occur beneath the surface of nano materials. For example, gate dielectrics in advanced semiconductors can have a physical thickness of less than one nanometre; the performance of these dielectrics can be predicted only by evaluating their equivalent electrical thickness. Similar considerations apply to carbon nanotubes, silicon wires and graphene—the basis for many nano innovations.

One of the main challenges in electrical characterisation of nano materials and structures is dealing with ultra-low signal levels. Another challenge is the wide range of behaviour that these materials and components can exhibit. For example, polymer materials can have resistances greater than one gigaohm. However, when drawn into fibres less than 100 nm in diameter and doped with various nanoparticles, a polymer may be changed from a superb insulator into a highly conductive wire. The result is an extremely wide range of test signals.

Detecting tiny electrical signals at the low end of the range requires high-sensitivity, high-resolution instruments such as electrometers, picoammeters and nanovoltmeters. Also, using these instruments for high-level signals as well demands instruments with a very wide dynamic range.

T&M manufacturers are also faced with the challenge of a shorter turnaround time. With the turnaround time (concept to design to volume production) increasingly becoming shorter due to the market competition, new and efficient methods of verifying the integrity and quality of handsets and similar products are required by all handset makers. With elements of design regularly being developed in different places, it is important to have common industry-standard test tools that enable groups to share information and resolve problems quickly. The quickening rate of technology changes means a faster cycle time for new prod-

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ucts entering the verification stage.

High-volume manufacturing requires innovations from T&M vendors so as to lower the test costs. Handset manufacturers demand solutions that can help them reach the market faster with a lower cost of testing and consequently a lower cost of production.

India: Still a costly affair

“There are already a few Indian T&M companies having their manufacturing base in India, while it is not the case for MNCs. In India, the import duty on raw materials is still high and therefore importing the finished goods is cheaper than locally producing them. As the Indian T&M market is still taking the shape and India is yet to be known as one of the best places for EMS, setting up manufacturing facility in India may be a costly affair for any global T&M maker,” says Sumit Sharma, marketing manager-India, Good Will Instrument.

Need of the hour

There are a few steps that can be taken to help India become the R&D hub for T&M equipment.

Naresh Narasimhan, country marketing manager, Tektronix India, says, “Companies are looking at world-class manufacturing and lean manufacturing techniques to reduce the production costs so that the benefits can be passed on to the end-user. Having said this, the infrastructure should be improved tremendously. Besides, the government should change its policies and automation should be considered. Components and sub-systems required for manufacturing, like chassis/power supply modules, should be made available at zero duty. Excise duty and sales tax for electronic test equipment (presently 8.24 and 12.5 per cent, respectively) should be brought down. Finally, more manufacturing units from support industries like plastic moulding units for quality cabinet design and manufacturing are needed.” ●

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